

Cable-Stayed Bridge over River Chenab

Tender Reference: TND-2026-00003 | Issued by: Punjab Roads & Infrastructure Authority (PRIA)

02 — Scope of Work

1. Hydrological & Geotechnical Studies

Contractor to conduct or verify hydrological studies for 100-year flood (Q100) discharge calculations. Geotechnical investigation: 8 boreholes minimum to 40 m depth, SPT testing, lab analysis. Report submission within 60 days of commencement.

2. Foundation Works

Main pylon foundations: large-diameter bored piles (1,500 mm dia.) to a minimum depth of 35 m below existing riverbed level. 24 piles per pylon, 2 pylons total. Pile caps: 8 m x 8 m x 3.5 m deep, Grade C-45 concrete. River crossing cofferdam installation and dewatering.

3. Pylon Construction

Two H-shaped reinforced concrete pylons, height 85 m above deck level. Pylon legs: 3.0 m x 2.5 m box section at base tapering to 2.0 m x 1.8 m at top. Concrete grade C-50, high-performance mix design. Pylon crossbeam at deck level and upper level.

4. Deck Construction

Composite steel-concrete deck system. Main girders: twin steel box girders (2,400 mm depth) with 250 mm reinforced concrete deck slab. Total deck width: 13.5 m (dual 3.5 m lanes, 1.5 m shoulders each side, 1.0 m median). Approach spans (2 x 150 m): precast prestressed concrete box girders.

5. Stay Cables

Fan cable arrangement: 32 pairs of high-strength parallel wire cables (PWS system), 7 mm dia. wires, breaking strength $\geq 1,860$ MPa. Cable anchorage system at pylon and deck level. PE pipe outer sheath with anti-vandalism coating.

6. Road Works

Approach roads: 800 m each end, flexible pavement (100 mm DBM + 50 mm AC wearing course). Road markings, crash barriers, lighting, and signage.

7. River Training & Protection

Stone pitching (600 mm thick) for 200 m upstream and downstream of each pylon for scour protection. Anti-scour mattresses at pile cap level.